

# Explainer: Problem 3 (Rudin 6.18) — Same Range, Different Lengths

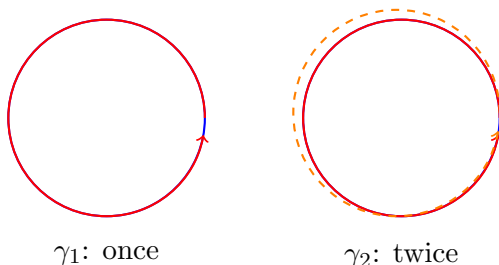
## The three curves

All three map  $[0, 2\pi]$  into the unit circle in  $\mathbb{C}$ :

|                                        |                                  |
|----------------------------------------|----------------------------------|
| $\gamma_1(t) = e^{it}$                 | (once around, counterclockwise)  |
| $\gamma_2(t) = e^{2it}$                | (twice around)                   |
| $\gamma_3(t) = e^{2\pi i t \sin(1/t)}$ | (wild oscillation near $t = 0$ ) |

## Same range, different curves

A **curve** is a *map*, not just its image. Two curves with identical image (the unit circle here) can have very different “lengths” because length counts *how many times* you traverse.  $\gamma_2$  goes around twice, so length  $= 4\pi = 2 \cdot (2\pi)$ .



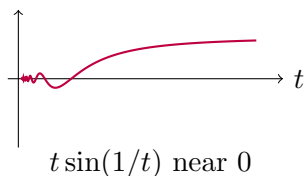
## Length of a curve

Length  $= \sup_P \sum_i |\gamma(t_i) - \gamma(t_{i-1})|$ , the supremum of polygonal lengths over partitions  $P$  of  $[a, b]$ . A curve is **rectifiable** if this sup is finite.

For smooth curves there's a formula: length  $= \int_a^b |\gamma'(t)| dt$ . Both  $\gamma_1$  and  $\gamma_2$  are smooth with  $|\gamma'_j| = j$ , giving  $j \cdot 2\pi$ .

## Why $\gamma_3$ is not rectifiable

Near  $t = 0$ , the function  $t \sin(1/t)$  oscillates wildly. The exponent  $2\pi i t \sin(1/t)$  makes  $\gamma_3$  sweep around the circle infinitely often in any neighborhood of  $t = 0$ , and these sweeps don't shrink fast enough.



**Strategy:** Find a sequence of times  $t_n \rightarrow 0$  where  $t_n \sin(1/t_n)$  hits half-integer values (so  $\gamma_3(t_n)$  alternates  $\pm 1$  on the circle). Then the polygonal sum  $\sum |\gamma_3(t_n) - \gamma_3(t_{n+1})|$  accumulates  $\geq 2$  each step, infinitely many times — diverges.

**Hint:** Try  $t_n$  such that  $1/t_n \sin(1/t_n) = \text{integer}$ , or pick  $t_n = 1/(n\pi/2)$  type values to make  $\sin(1/t_n) = \pm 1$  and catch the extremes of the oscillation.